PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Most Likely - Ranked Follow-up - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions in the ranked study order. These continue the shortlist after the main test; this longer paper is extra practice, not the announced exam format.

Write all rough calculations and final answers on separate paper. Show your working.

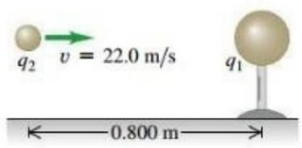
Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.5

23.5 •• A small metal sphere, carrying a net charge of $q_1 = -2.80\ \mu C$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80\ \mu C$ and mass 1.50 g, is projected toward q_1. When the two spheres are 0.800 m apart, q_2, is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1?

Figure **E23.5**

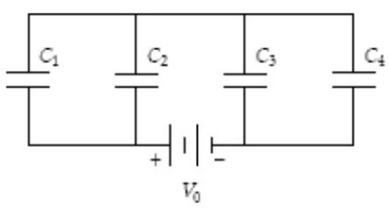


2. [____] Tutorial 6 - Question 69

69. A parallel-plate capacitor of plate separation d is charged to a potential difference ΔV_0. A dielectric slab of thickness d and dielectric constant κ is introduced between the plates while the battery remains connected to the plates. (a) Show that the ratio of energy stored after the dielectric is introduced to the energy stored in the empty capacitor is $U/U_0 = \kappa$. Give a physical explanation for this increase in stored energy. (b) What happens to the charge on the capacitor? (Note that this situation is not the same as in Example 26.7, in which the battery was removed from the circuit before the dielectric was introduced.)

3. [____] Tutorial 7 - Question 1

1. What is the total energy stored by C_3 when $C_1 = 50\ \mu F$, $C_2 = 30\ \mu F$, $C_3 = 36\ \mu F$, $C_4 = 12\ \mu F$, and $V_0 = 30$ V?



a. 6.3 mJ
b. 25 mJ
c. 57 mJ
d. 1.6 mJ
e. 14 mJ

4. [____] Tutorial 4 - Question 7

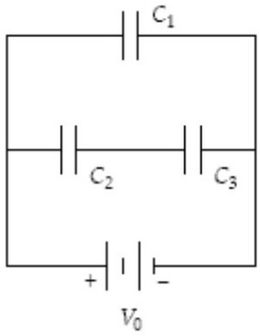
7. A 4.0-pC point charge is placed at the center of a hollow (inner radius = 2.0 cm, outer radius = 4.0 cm) conducting sphere which has a net charge of 4.0 pC. Determine the magnitude of the electric field at a point which is 6.0 cm from the point charge.
   a. 35 N/C
   b. 25 N/C
   c. 30 N/C
   d. 20 N/C
   e. 10 N/C

5. [____] Tutorial 4 - Question 9

9. A point charge of 6.0 nC is placed at the center of a hollow spherical conductor (inner radius = 1.0 cm, outer radius = 2.0 cm) which has a net charge of –4.0 nC. Determine the resulting charge density on the inner surface of the conducting sphere.
   a. +4.8 μC/m^2
   b. –4.8 μC/m^2
   c. –9.5 μC/m^2
   d. +9.5 μC/m^2
   e. –8.0 μC/m^2

6. [____] Tutorial 7 - Question 2

2. What is the potential difference across C_2 when $C_1 = 5.0\ \mu$F, $C_2 = 15\ \mu$F, $C_3 = 30\ \mu$F, and $V_0 = 24$ V?



   a. 21 V
   b. 19 V
   c. 16 V
   d. 24 V
   e. 8.0 V

7. [____] Tutorial 7 - Question 7

7. A parallel plate capacitor of capacitance C_0 has plates of area A with separation d between them. When it is connected to a battery of voltage V_0, it has charge of magnitude Q_0 on its plates. It is then disconnected from the battery and the space between the plates is filled with a material of dielectric constant 3. After the dielectric is added, the magnitudes of the charge on the plates and the potential difference between them are
   a. $\frac{1}{3}Q_0, \frac{1}{3}V_0$.
   b. $Q_0, \frac{1}{3}V_0$.
   c. Q_0, V_0.
   d. $Q_0, 3V_0$.
   e. $3Q_0, 3V_0$.

8. [____] Homework 7 - Question 24.36

24.36 • A budding electronics hobbyist wants to make a simple 1.0-nF capacitor for tuning her crystal radio, using two sheets of aluminum foil as plates, with a few sheets of paper between them as a dielectric. The paper has a dielectric constant of 3.0, and the thickness of one sheet of it is 0.20 mm. (a) If the sheets of paper measure 22 × 28 cm and she cuts the aluminum foil to the same dimensions, how many sheets of paper should she use between her plates to get the proper capacitance? (b) Suppose for convenience she wants to use a single sheet of posterboard, with the same dielectric constant but a thickness of 12.0 mm, instead of the paper. What area of aluminum foil will she need for her plates to get her 1.0 nF of capacitance? (c) Suppose she goes high-tech and finds a sheet of Teflon of the same thickness as the posterboard to use as a dielectric. Will she need a larger or smaller area of Teflon than of posterboard? Explain.